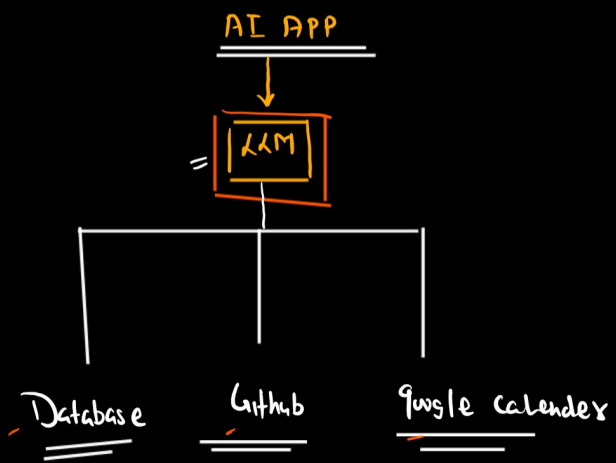
MCP

Normal AI app:-

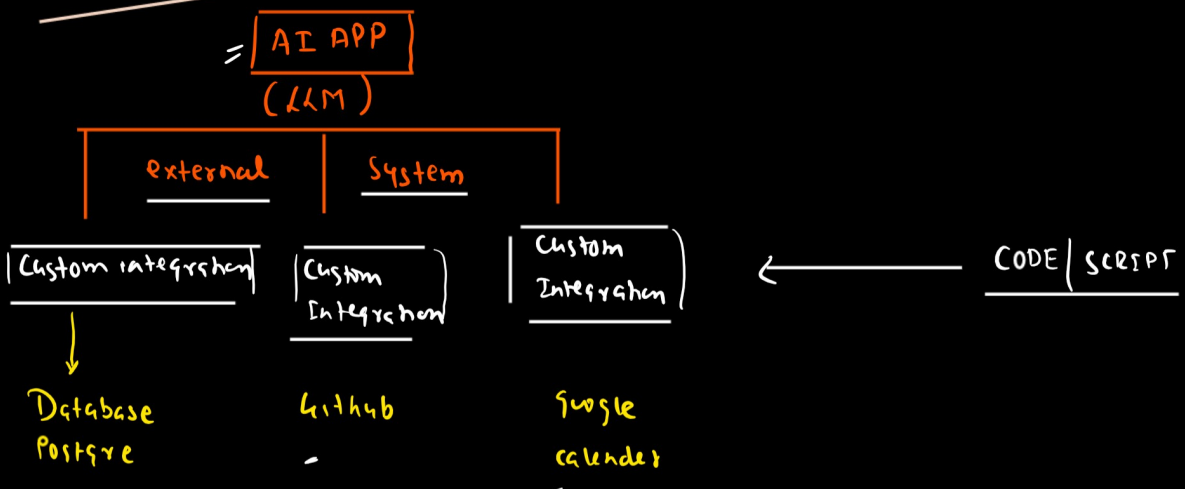


What if LLM needs info from external system (AI APP)



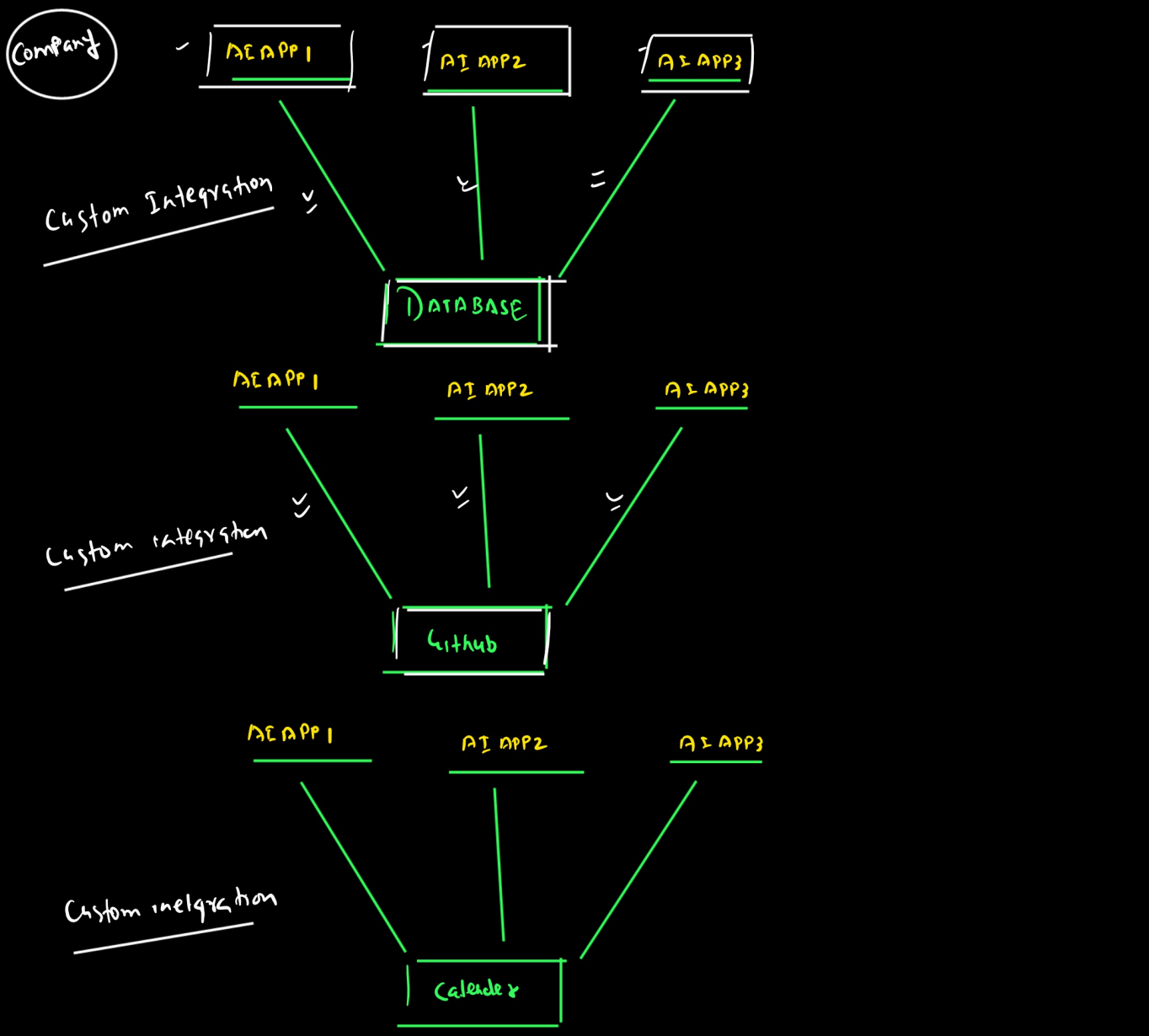
LLM itself cannot magically access all these systems. We need integrations.

Without mcp

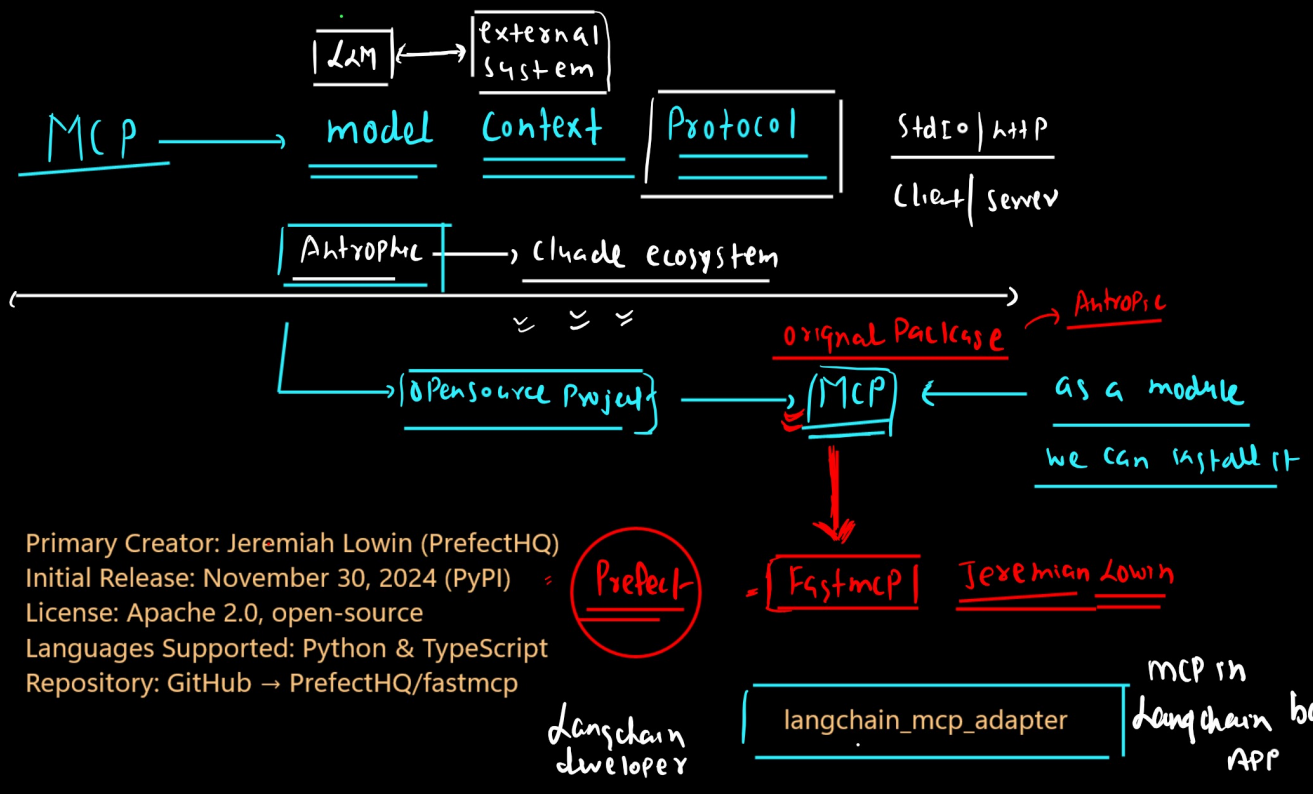


Without MCP if my AI application wants to communicate with five different systems, I may have to create and maintain different integrations for each system.

- AI → (Database integration) ✓
- AI → (GitHub integration) ✓
- AI → Slack integration ✓
- AI → Calendar integration ✓
- AI → Filesystem integration ✓



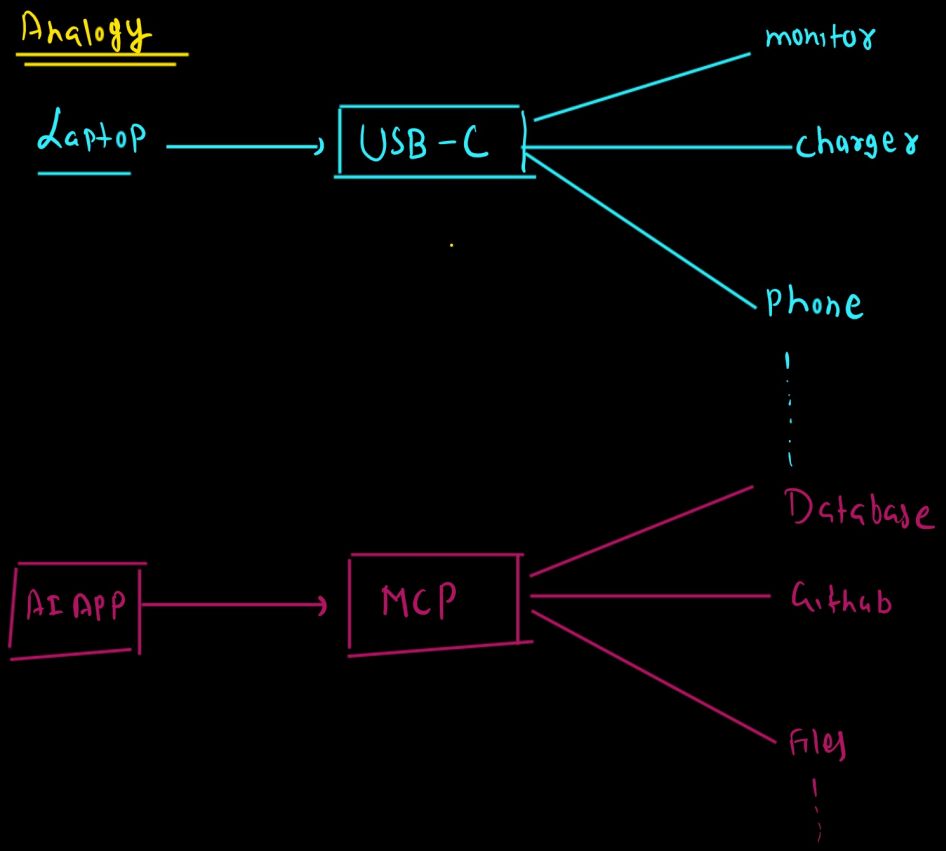
Without MCP = many custom AI-to-system integrations.



Term	What is it?	Purpose
MCP	Protocol/standard	Defines how AI apps and MCP servers communicate
MCP Python SDK	Official Python implementation	Build MCP clients/servers
MCPServer	Official SDK high-level server class	Expose tools/resources/prompts
FastMCP	Separate higher-level MCP framework	Build MCP apps more conveniently
LangChain MCP	LangChain integration/adapter	Let LangChain agents use MCP servers

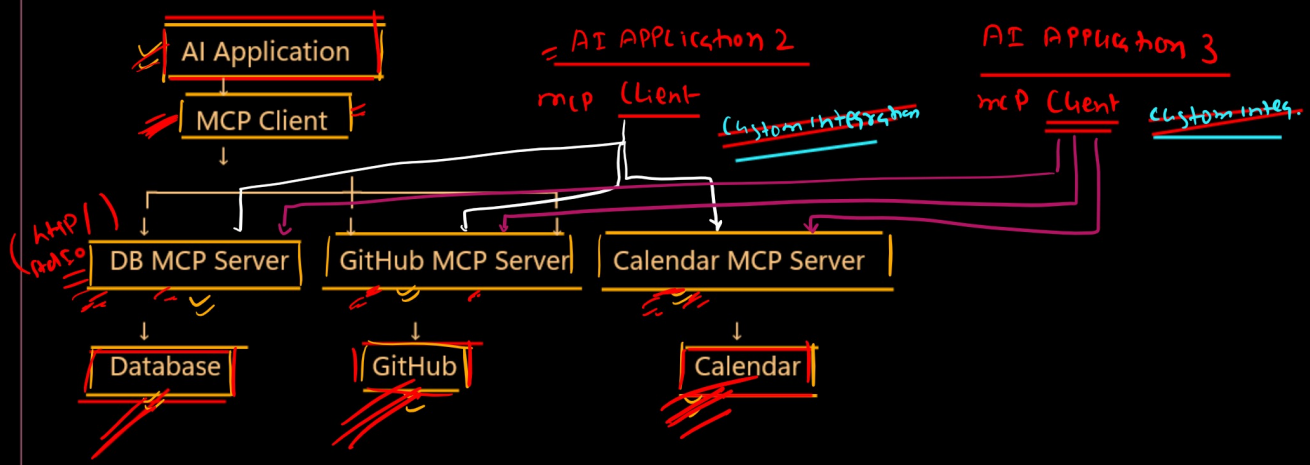
MCP is basically a standard connection system for AI applications.

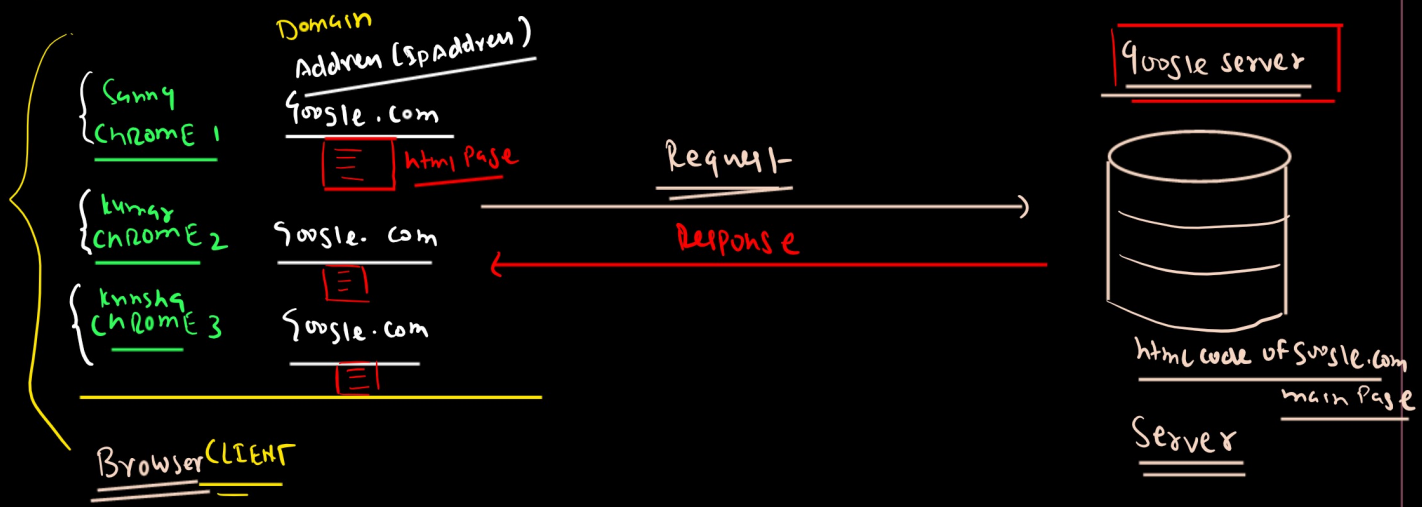
MCP, or Model Context Protocol, is a standard protocol that allows AI applications to connect with external system.



USB-C standardizes how electronic devices connect. MCP standardizes how AI applications connect to external capabilities

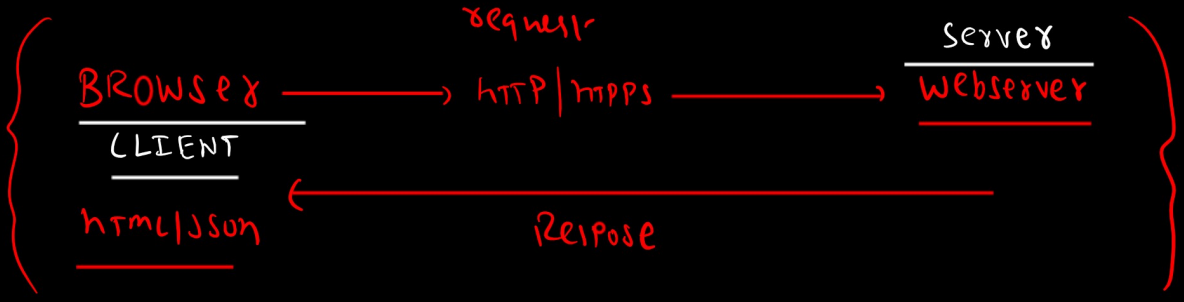
AI Application
- Custom code → Database
- Custom code → GitHub
- Custom code → Calendar
- Custom code → Files

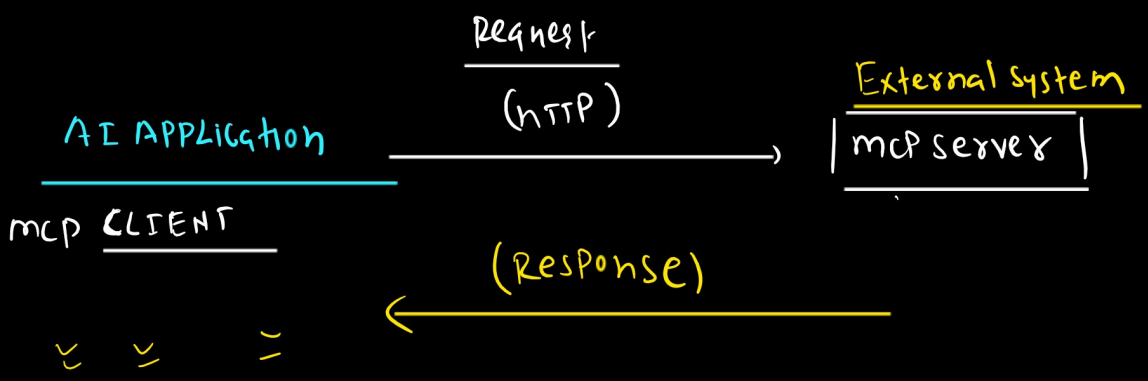


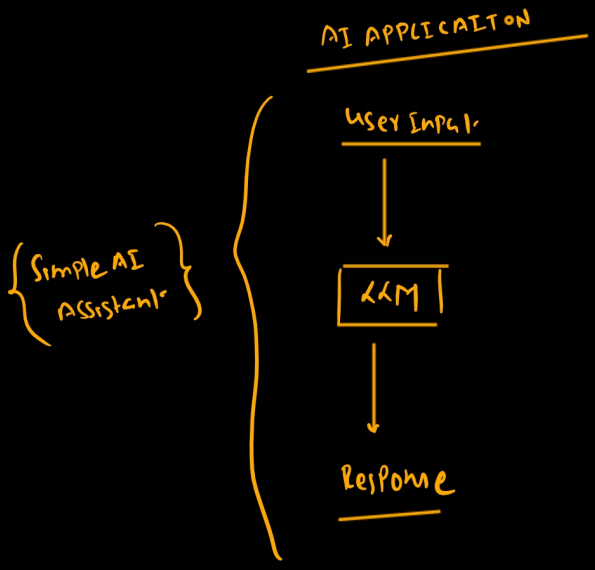


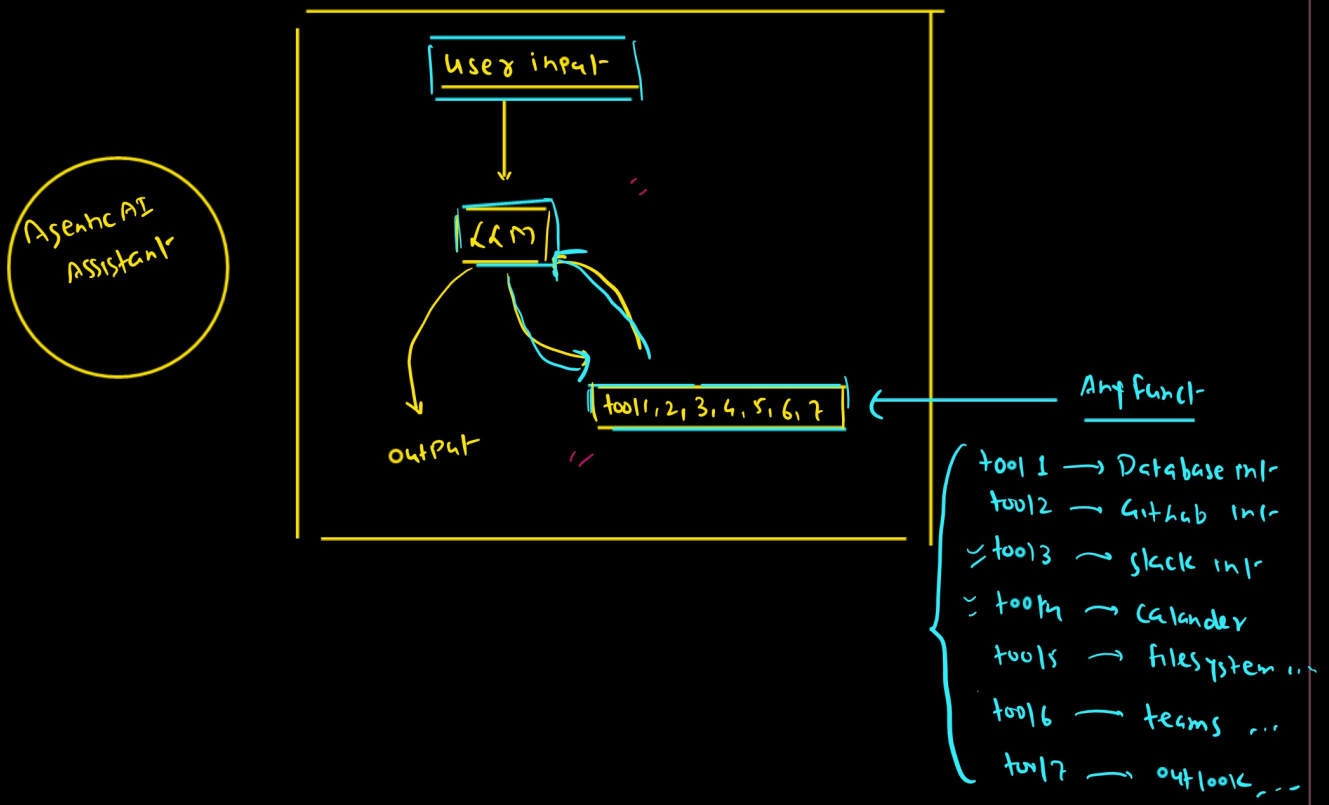
Without MCP, integration logic is usually custom. With MCP, AI applications and external capabilities can communicate using a common protocol.

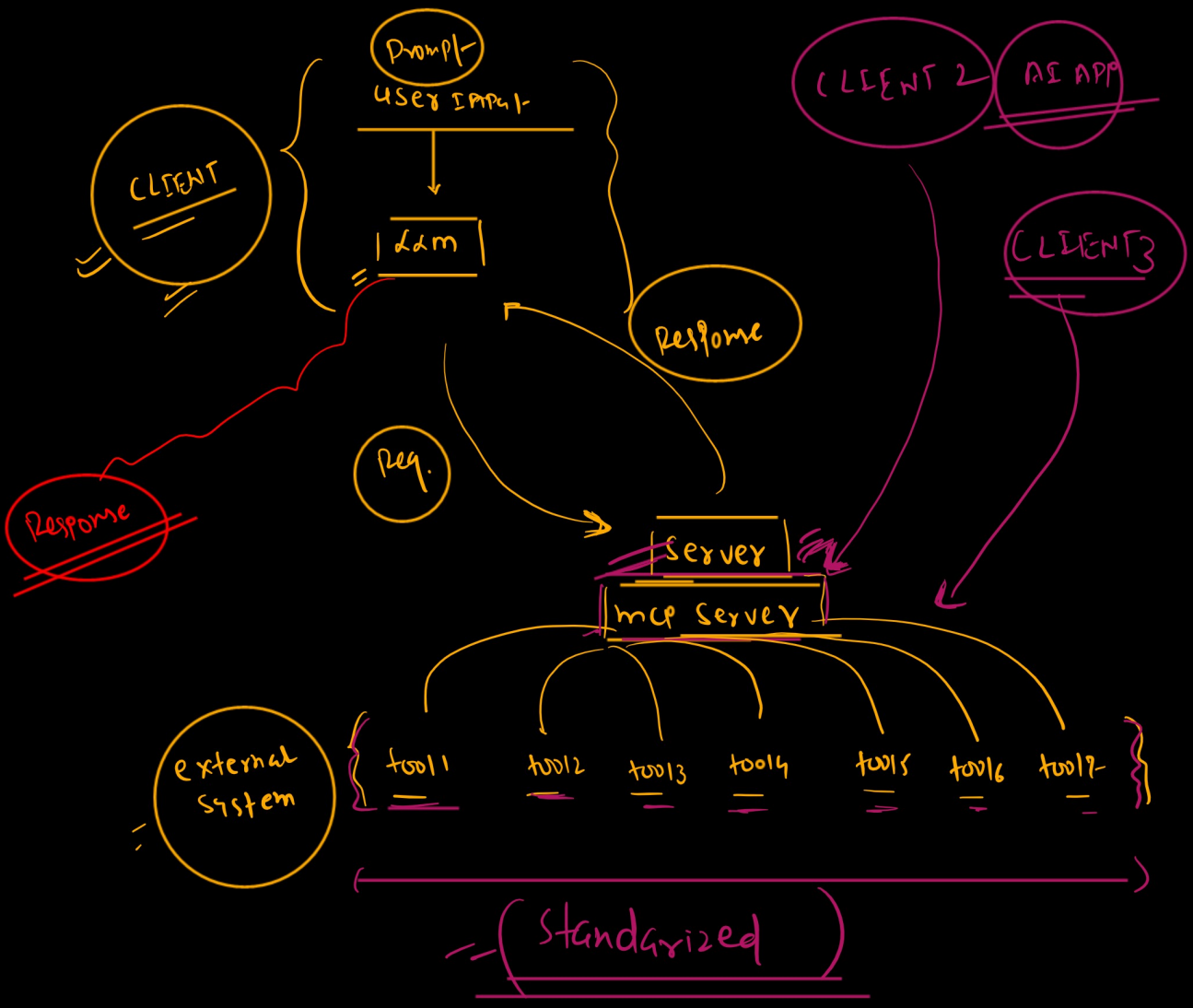


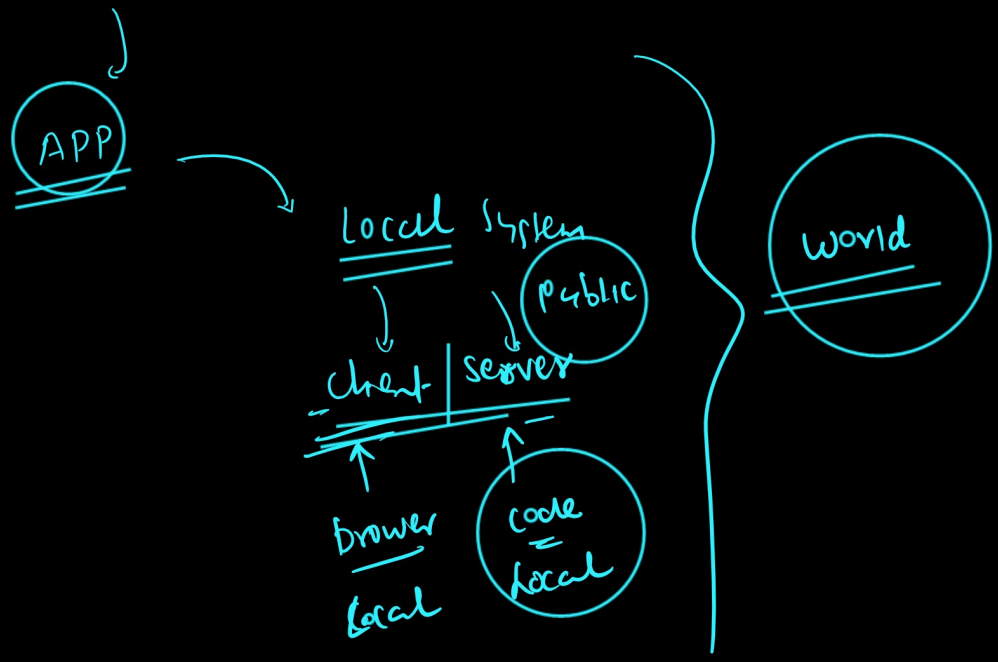


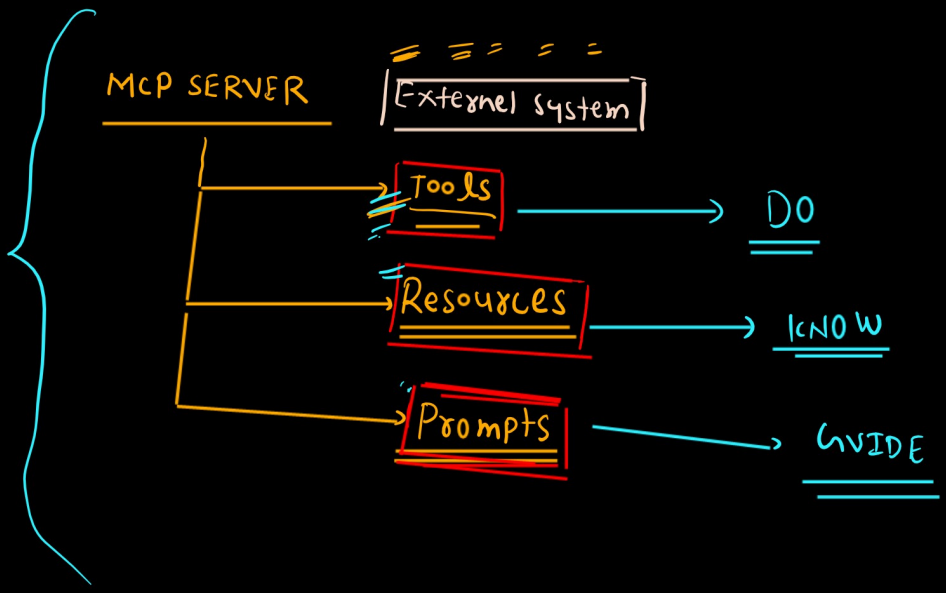

Request
(HTTP)
AI APPLICATION
MCP CLIENT
External system
MCP server
(Response)


AI APPLICATION
User Input
LLM
Response
Simple AI Assistant


User input
LLM
tool 1, 2, 3, 4, 5, 6, 7
output
Agentic AI Assistant
Any Func-
tool 1 → Database int-
tool 2 → Github int-
tool 3 → Slack int-
tool 4 → Calander
tool 5 → filesystem ...
tool 6 → teams ...
tool 7 → outlook ...


Prompt-
User Input-
CLIENT
Llm
Response
CLIENT 2
AI APP
CLIENT3
Response
Req.
Server
mcp Server
external system
tool1
tool2
tool3
tool4
tool5
tool6
tool7-
Standarized


APP
Local System
public
Client
Server
Brower
Local
code
Local
World



TRAVEL AGENCY (AI system)

TOOL → executable function

- search_flight() @tool
- book_hotel() @tool
- book_cab() @tool
- search_restaurant() @tool

User:

Find flight from BLR to Dubai?

Resources:

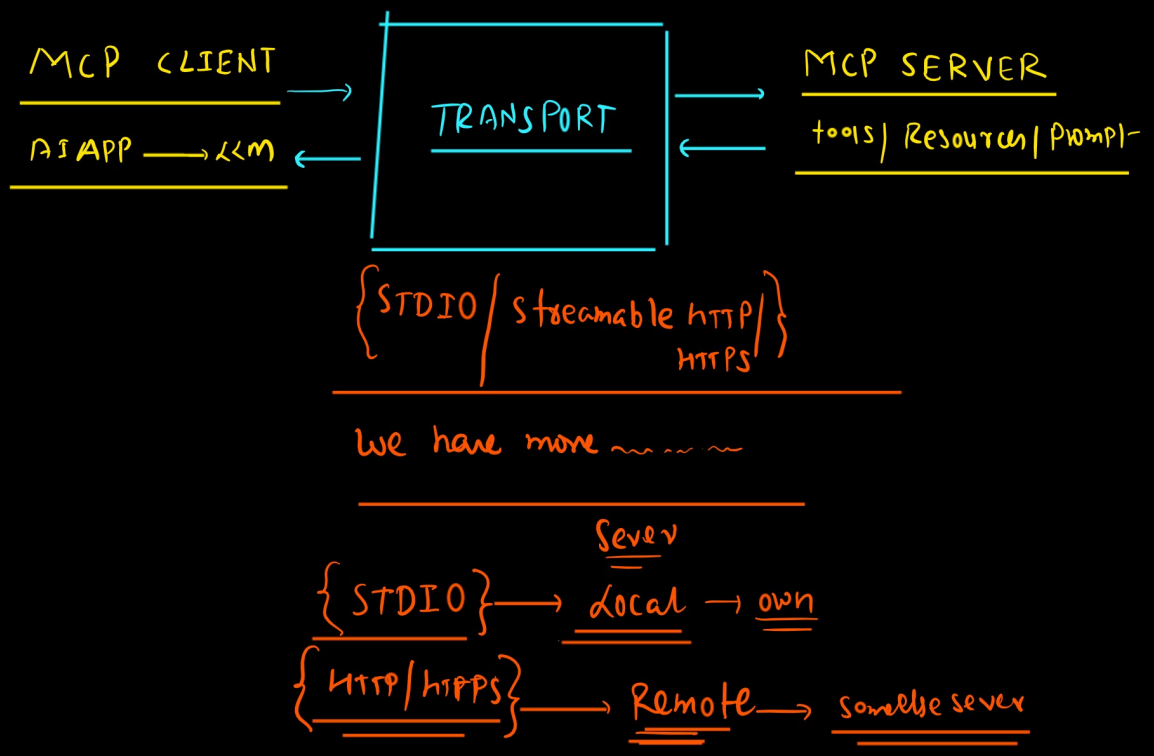
Resource is information that the application can read and provide to the model as context

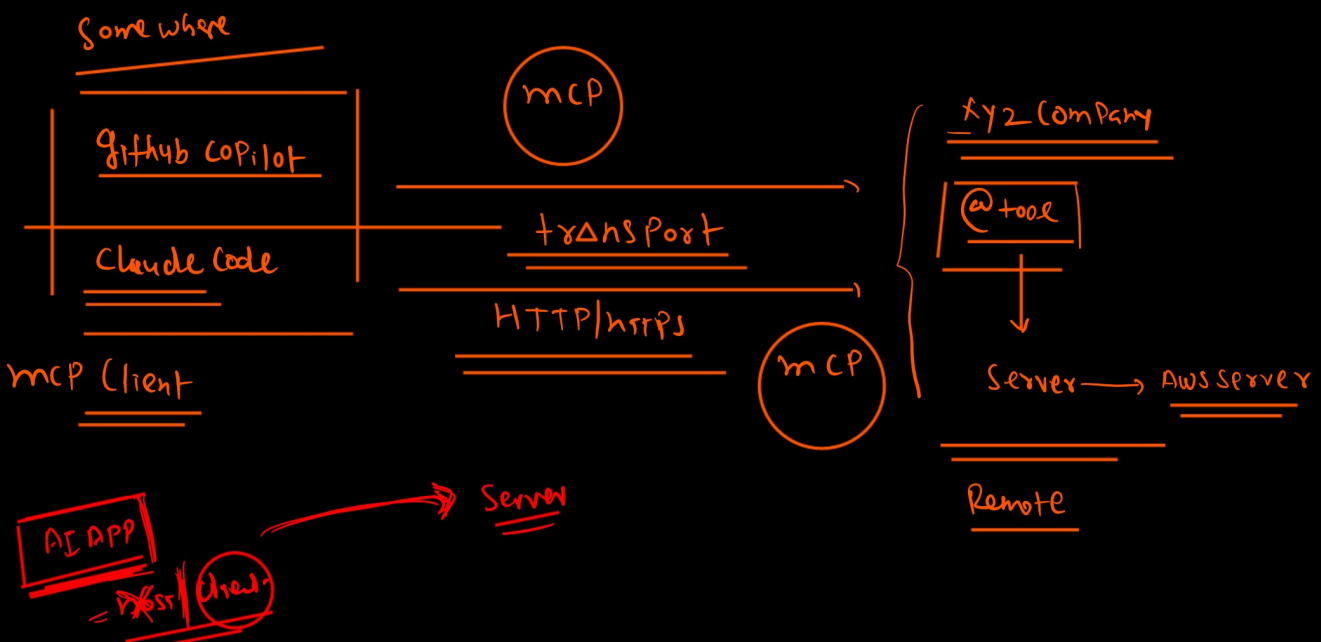
travel://preferences
company://policy
calendar://events

Prompt:-

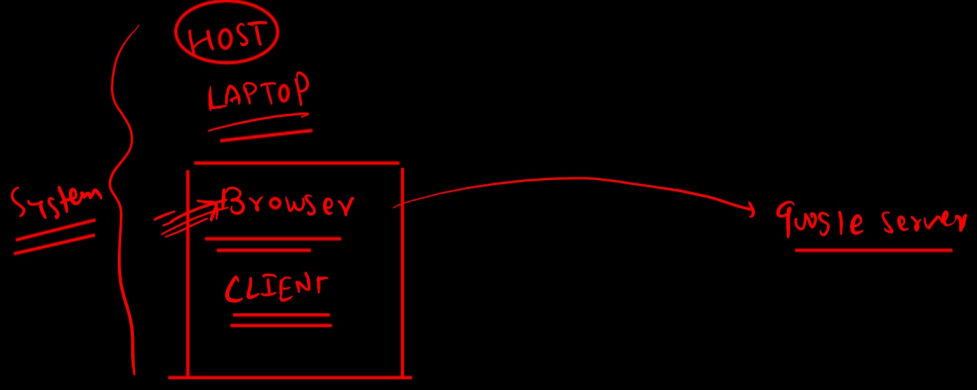
Prompt is a reusable instruction template

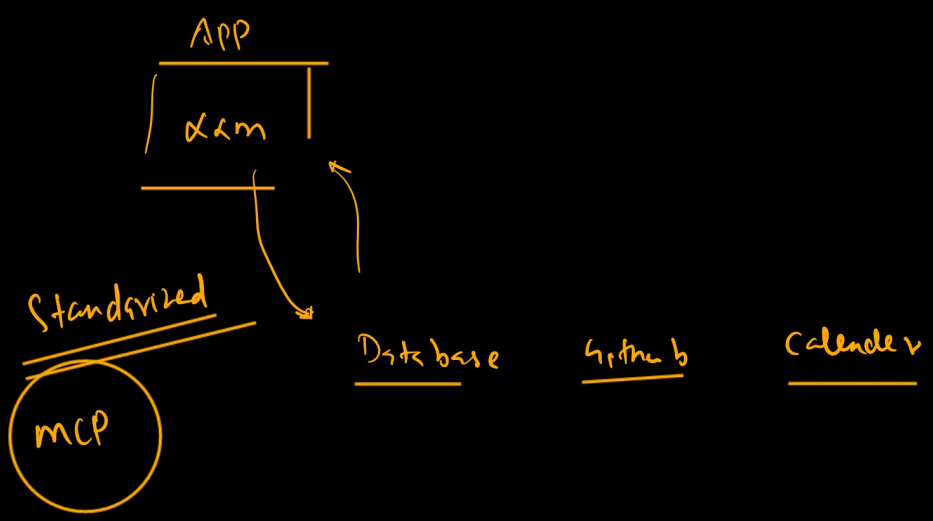
- Plan vacation
- book flight





MCP is a standard protocol for connecting AI applications to external systems. The AI application acts as the MCP Host. It uses MCP Clients to connect to MCP Servers. MCP Servers expose Tools for actions, Resources for data, and Prompts for reusable instructions. If the server is local, communication can happen through STDIO; if it is remote, Streamable HTTP can be used. MCP does not replace APIs, LLMs, databases or agents—it standardizes how AI applications interact with external capabilities.




App
Llm
Standarized
mcp
Database
Github
Calender